REPUBLIC OF TURKEY
YILDIZ TECHNICAL UNIVERSITY
DEPARTMENT OF COMPUTER ENGINEERING

CLASSIFICATION OF HYPERSPECTRAL IMAGES USING MACHINE LEARNING AND DEEP LEARNING METHODS

22011083 — TAN ERCİYAS
22011084 — KEREM BAYDAR

SENIOR PROJECT

Advisor
Dr. Ahmet ELBİR

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TAN ERCİYAS
KEREM BAYDAR

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LIST OF ABBREVIATIONS

1D-CNN	1-Dimensional Convolutional Neural Network
3D-CNN	3-Dimensional Convolutional Neural Network
AE	Autoencoder
R&D	Research and Development
CNN	Convolutional Neural Networks
GAN	Generative Adversarial Networks
GPU	Graphics Processing Unit
GUI	Graphical User Interface
HSI	Hyperspectral Images
MRF	Markov Random Fields
PCA	Principal Component Analysis
RF	Random Forest
RNN	Recurrent Neural Networks
SAE	Stacked Autoencoders
SVM	Support Vector Machines
ViT	Vision Transformer

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ABSTRACT

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TAN ERCİYAS
KEREM BAYDAR

Department of Computer Engineering
Senior Project

Advisor: Dr. Ahmet ELBİR

BU KISIM İLERİDE DOLDURULACAKTIR

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ÖZET

HİPERSPEKTRAL GÖRÜNTÜLERİN MAKİNE ÖĞRENMESİ VE DERİN ÖĞRENME YÖNTEMLERİYLE SINIFLANDIRILMASI

TAN ERCİYAS
KEREM BAYDAR

Bilgisayar Mühendisliği Bölümü
Bitirme Projesi

Danışman: Dr. Ahmet ELBİR

BU KISIM İLERİDE DOLDURULACAKTIR

Anahtar Kelimeler: BU KISIM İLERİDE DOLDURULACAKTIR

1
INTRODUCTION

Hyperspectral images (HSI) are rich data structures collected over a wide range of the electromagnetic spectrum, reflecting the physical and chemical properties of objects on the Earth's surface with high precision. Unlike traditional images, they contain hundreds of narrow spectral bands in each pixel, making them extremely valuable in fields such as agriculture, environmental monitoring, and the defense industry. However, this rich information leads to the problem known as the "curse of dimensionality" and results in significantly high computational costs during the training of machine learning and deep learning models. In addition, the number of labeled samples in hyperspectral datasets is generally very limited, which severely restricts the generalization capability of the developed classification models.

The motivation of this project is to conduct an in-depth investigation into the performance potential of supervised and semi-supervised deep learning architectures, such as CNN, ViT, and Self-Training, against the common challenges of high dimensionality and labeled data scarcity in hyperspectral image classification. The most significant original contribution of the project is the integration of the Curriculum Learning strategy to increase training stability and the development of a user-friendly Graphical User Interface (GUI) operating on a multithreaded architecture. This structure allows the complex models to be testable in real-time by the end-user.

Within the scope of this report; Chapter 2 presents a literature analysis and an evaluation of previous studies related to the project topic. Chapter 3 explains the system design and methodology, extending from data preprocessing to model architectures. Chapter 4 analyzes the obtained experimental results comparatively using various performance metrics, and Chapter 5 discusses the project results and provides suggestions for future work.

1.1 Purpose

The primary objective of this project is to quantitatively demonstrate the performance differences between supervised and semi-supervised learning approaches in the classification of hyperspectral images with high-dimensional and complex spectral data. In line with this main goal, the sub-objectives are as follows:

- To analyze model performances on three different fundamental datasets that are considered standards in the literature: Indian Pines, Pavia University, and Salinas.
- To examine the effects of data preprocessing steps, such as Gaussian filtering for noise reduction and Principal Component Analysis (PCA) for dimensionality reduction, on classification success.
- To integrate the Curriculum Learning strategy into the models to maximize stability during the training processes.
- To conduct an objective comparison of the models using Accuracy, Precision, Recall, F1-Score, and Kappa metrics.
- To develop a multithreaded Graphical User Interface (GUI) based on React and Python to prevent the developed models from being confined to desktop or laboratory environments and to demonstrate real-time prediction capabilities.

1.2 Preliminary Investigation

Preliminary investigations conducted during the study of the project area show that traditional machine learning methods are rapidly giving way to deep learning-based approaches in the hyperspectral image classification literature. Research into the gaps of previous studies reveals that existing systems can generally only produce successful results with large amounts of labeled data. However, in real-world scenarios, the labeling of pixels by experts is extremely costly and time-consuming.

The path to be followed in this project targets this gap in the literature by demonstrating how semi-supervised algorithms can overcome this bottleneck by utilizing a small amount of labeled data and a large amount of unlabeled data together. Furthermore, another issue identified in current systems is that models developed by researchers are usually only testable at the code level. This project directly addresses the need for converting theoretical successes into a practical and user-oriented application by introducing an interface that runs multithreaded model instances, enabling users to load external data and receive instantaneous predictions.

2
LITERATURE ANALYSIS

The problem of hyperspectral image (HSI) classification has been one of the most researched topics in the remote sensing literature for many years, aiming to increase object recognition sensitivity by utilizing rich spectral data content. In this process, which extends from traditional machine learning algorithms to modern deep learning-based architectures, many different methods have been proposed, and solutions have been sought for the "curse of dimensionality" and limited labeled data problems. In this section, the approaches in the literature are examined in a comparative manner, grouped according to their algorithmic foundations.

2.1 Traditional Machine Learning Approaches

Early studies in the classification of hyperspectral images predominantly relied on statistical and traditional machine learning methods such as Support Vector Machines (SVM) and Random Forest (RF). Du et al. achieved high accuracy rates (97.71% for Indian Pines) using discriminative sparse representations and multinomial logistic regression [1]. Similarly, Xia et al. attempted to optimize the performance of RF models by applying extended profiles and ensemble strategies [2]. Furthermore, cascaded random forest (Cascaded RF) architectures, where multiple classifiers are trained simultaneously, were proposed to prevent overfitting [3]. Among SVM-based approaches, studies combining probabilistic SVMs with guided filters [4] and integrating spectral-spatial information with Markov Random Fields (MRF) [5] stand out. Although these methods are relatively resistant to noise and offer interpretability, they have fallen behind deep learning algorithms in terms of performance due to their difficulties in dealing with high-dimensional data and the necessity of extracting complex spatial features manually (hand-crafted).

2.2 Convolutional Neural Networks (CNN) Based Approaches

The capacity of Convolutional Neural Networks (CNNs) to process both spatial and spectral information simultaneously has made them a standard in the literature. Chen et al. aimed to solve the high dimensionality problem using 3D Convolutional Neural Networks (3D-CNN) and achieved 99.66% success on the Pavia University dataset with a model supported by L2 norm and dropout techniques [6]. Li et al. proposed a 1D-CNN structure that generates deep pixel-pair features to overcome the limited training data problem [7]. To reduce the high number of parameters and computational costs of 3D-CNN models, various architectures have been developed, such as HybridSN [8], which combines 3D and 2D convolution layers; fast and compact 3D-CNN architectures [9]; and next-generation networks that prevent data loss with border mirroring techniques [10]. Although these methods successfully extract spectral-spatial features, the requirement for large amounts of labeled data and high GPU memory costs as model depth increases remain their primary disadvantages.

2.3 Autoencoders, Generative Networks, and RNNs

The requirements for dimensionality reduction and dealing with limited labeled samples have encouraged the use of Autoencoders (AE), Generative Adversarial Networks (GANs), and Recurrent Neural Networks (RNNs) in the literature. Li et al. proposed a method combining Stacked Autoencoders (SAE) with CNNs to adaptively weight local spatial context [11]. Similarly, Zhao et al. integrated SAEs for dimensionality reduction with 3D deep residual networks [12]. RNNs, which are successful in time series, were adapted by Shi et al. to fuse multi-scale spectral-spatial features [13]. On the other hand, GANs have been utilized to artificially augment limited training data with synthetic samples in an attempt to increase the generalization capability of models [14]. Although these architectures improve accuracy in certain areas, their practical use has been limited by scalability issues arising from the sequential nature of processing and the highly unstable training processes of generative networks.

2.4 Transformer Architectures and Semi-Supervised Methods

The Transformer architecture, one of the most innovative developments in the field of deep learning in recent years, has begun to be used in HSI classification due to its ability to model long-range dependencies. Hybrid models combining the local feature extraction capabilities of CNNs with the global attention mechanisms of Transformers [15, 16], systems based on spectral-spatial feature tokenization [17], and dual attention networks [18] have come to the fore. Although these methods achieve over 99% success on various datasets, they require extreme computational power compared to traditional machine learning and CNN methods.

Another successful alternative against labeled data scarcity is the semi-supervised 3D deep neural networks examined by Sellami et al. [19]. This study demonstrated the potential of incorporating unlabeled data into the process with adaptive band selection. However, the risk of bias created by these systems while drawing decision boundaries and the difficulty of selecting optimum semi-supervised threshold values remain ongoing problems.

2.5 Current Standing in Literature and Original Value of the Study

Considering the literature reviewed, it is clearly observed that Transformer and deep CNN-based models achieving high accuracy are prone to overfitting under limited labeled data conditions and demand high hardware resources. Although semi-supervised learning and generative models attempt to offer solutions to these hardware and data bottlenecks, the lack of integrated strategies that guarantee model training stability is noteworthy.

In this direction, our project aims to gather scattered algorithmic advantages into a single semi-supervised framework by combining CNN and Transformer-based architectures with Self-Training strategies and the "Curriculum Learning" strategy, which will address the unbalanced learning problems of models in the literature. Unlike complex models that are generally left as mere Python scripts or laboratory codes, our project will provide a user-friendly Graphical User Interface (GUI) based on React and Python capable of simultaneous inference (multithreaded inference). In this way, the high accuracy potential identified in the literature will be transformed from a purely academic result into a testable and practical structure.

3
SYSTEM ANALYSIS AND FEASIBILITY

The purpose of system analysis is to identify and define the main elements and functions to find the most appropriate solution for the project. In this section, the objectives of the project are detailed, and information sources and requirements are determined. By the end of this chapter, all requirements related to the project are defined, and the most suitable solution for proceeding to the design phase is established.

The research and information gathering methods used for eliciting requirements and their results are provided in this section. In light of the determined requirements, the modules, users, and roles in the system are defined. System analysis also reveals the structure to test whether the needs are met upon project completion. For this reason, the performance metrics (speed, recognition success, etc.) to be used in the project are also defined in this section.

3.1 System Analysis

Due to the R&D nature of this project, the fundamental requirements were determined during the system analysis phase by examining the performance bottlenecks of existing deep learning-based hyperspectral classification models. The main function of the system is to process high-dimensional spectral data—either uploaded externally or selected from standard literature datasets such as Indian Pines [20], Pavia University [21], and Salinas [22]—to predict the class of each pixel with high accuracy.

The fundamental requirements that the system is expected to fulfill are as follows:

- De-noising of hyperspectral images using Gaussian filtering and dimensionality reduction through the Principal Component Analysis (PCA) method [23].
- Enabling the training of supervised (CNN [6], ViT [24]) and semi-supervised (Self-Training [25]) models using a Curriculum Learning [26] strategy.

- Providing a graphical user interface (GUI) operating on a multithreaded architecture to allow simultaneous testing and real-time prediction of developed models [27].

To test whether the requirements are met at the end of the project, the following performance metrics will be utilized: Accuracy, Precision, Recall, F1-Score, and Kappa coefficient.

3.2 Feasibility

The feasibility study evaluates the viability of the project by analyzing how resources were selected among alternatives and the project's timeline.

3.2.1 Technical Feasibility

Technical feasibility is a brand-independent study that highlights technical features, including software and hardware feasibility. Python [28] has been selected as the application development language due to its robust ecosystem in the field of machine learning. For deep learning processes, PyTorch [29] or TensorFlow [30] libraries will be used, while Scikit-learn [23] and OpenCV [31] will be utilized for data processing. For the front-end development of the GUI, React [27] has been preferred for its performance and concurrent processing capabilities. The development environment will be VS Code [32], and version control will be managed via GitHub [33].

3.2.1.1 Hardware Feasibility

The high parallel processing capacity required for training deep learning models will be met using the GPU accelerators provided by the Google Colab [34] environment. For front-end development and interface integration, the hardware owned by the project team (macOS-based computers) possesses sufficient capacity.

3.2.2 Legal Feasibility

The tools used in the project, such as Python, React, PyTorch, and OpenCV, have open-source licenses. The Indian Pines, Pavia University, and Salinas datasets are open for academic use. Therefore, it has been determined that the project does not pose any legal barriers or patent infringements.

3.2.3 Economic Feasibility

All libraries and development platforms selected for the project consist of free and open-source tools. Thanks to the use of Google Colab and existing personal hardware, no additional investment costs are incurred, and the project is considered economically viable.

3.2.4 Workforce and Time Feasibility

The project is conducted by Tan Erciyas and Kerem Baydar. The technical expertise of the team covers the necessary domains for project completion. The following Gantt chart illustrates the project timeline from February 2026 to June 2026.

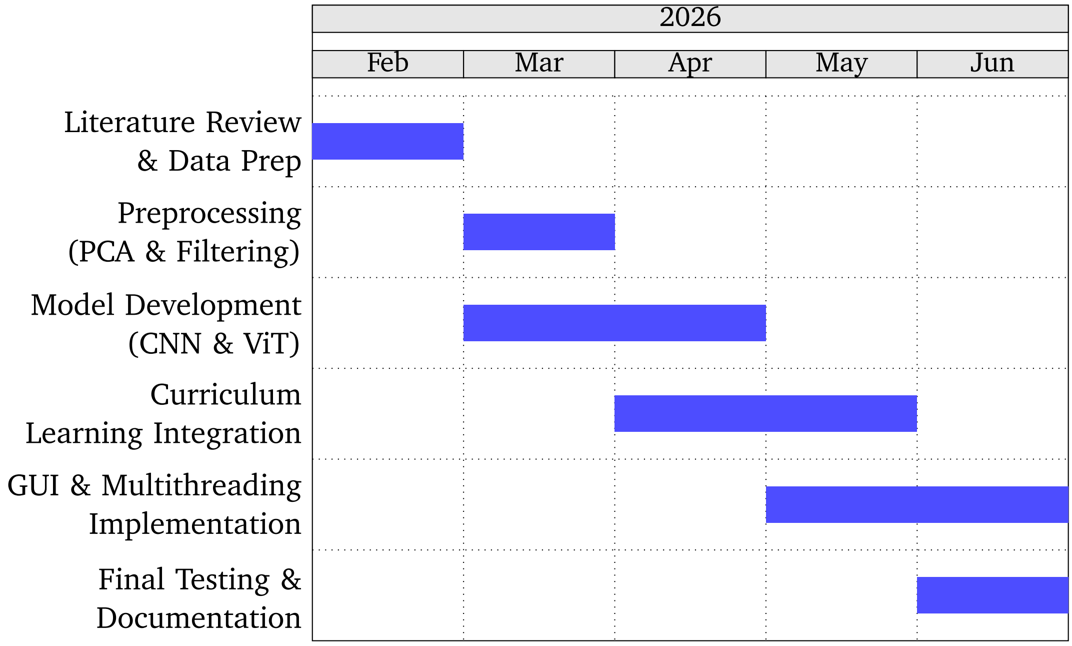

Figure 3.1 Project Gantt Chart (Feb 2026 - June 2026)

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Curriculum Vitae

FIRST MEMBER

Name-Surname: TAN ERCİYAS
Birthdate and Place of Birth: 15.12.2003, Trabzon
E-mail: tan.erciyas@std.yildiz.edu.tr
Phone: 0505 353 60 58
Practical Training: Genarion Software Department
Baykar Web Software Department
Tüpraş Software Department

SECOND MEMBER

Name-Surname: KEREM BAYDAR
Birthdate and Place of Birth: 06.11.2004, Kocaeli
E-mail: kerem.baydar@std.yildiz.tr
Phone: 0544 735 41 00
Practical Training: Baykar OOP Software Department
IBTECH Dw & Data Insights Department
Genarion Software Department
Uzer Makina IT Department

Project System Informations

System and Software: Windows İşletim Sistemi, Python
Required RAM: 2GB
Required Disk: 256MB